INFORMATICS INSTITUTE OF TECHNOLOGY

In Collaboration with

ROBERT GORDON UNIVERSITY ABERDEEN

A Multimodal Approach to Solar PV Fault Localisation and Rectification

Group 22 Project Proposal Document by

Tamadhi Liyanage - 20241004

Mohamed Razik Seyed Rumaiz - 20241705

R. M. Manuli Maneka Gokarella- 20232954

Dion Rasmika - 20241835

Supervised by

Ms. Suvetha Suvendran

Submitted in partial fulfilment of the requirements for the BSc (Hons) in Artificial Intelligence and Data Science degree at the Robert Gordon University.

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Declaration

We hereby certify that this project proposal and all the artifacts associated with it is our own work and it has not been submitted before nor is currently being submitted for any degree program.

Student Name	Student ID	Signature
Tamadhi Liyanage	20241004	
Mohamed Razik Seyed Rumaiz	20241705	
R. M. Manuli Maneka Gokarella	20232954	
Dion Rasmika	20241835	

I have read the project proposal and it is in accordance with the approved university project proposal outline.

Suvetha Suvendran
suvetha.s@iit.ac.lk
Signature of the supervisor:

Date: 26/10/2025

Contents

1 Introduction

This project aims to develop an AI-driven system for the automated management of faults in solar photovoltaic (PV) systems. Our system utilizes electrical and image data to identify the faults, identifies its location, assess the severity of the fault, and finally recommend appropriate rectification actions. By integrating these four components, this system enhances the reliability of solar PV operations by minimizing the diagnostic time, reducing the downtime, and preventing energy yield losses.

2 Problem Domain

The global adoption of solar PV systems has been rising rapidly, driven by declining panel costs and supportive government policies. According to Fraunhofer (2019), PV installations worldwide grew at a Compound Annual Growth Rate (CAGR) of around 24% between 2010 and 2017.

However, despite this rapid growth, large-scale PV systems continue to face significant performance challenges. Underperformance in solar farms is not a minor issue but a systemic one, primarily driven by electrical faults at the system level. Crucially, string-level anomalies account for about 23% of total observed power losses across large PV plants worldwide, making it the second-largest contributor after inverter issues (Raptor Maps, 2025).

Recent inspection data shows how severe performance issues are at the system level. A case study revealed that both system and module-level equipment anomalies caused a 3.7% reduction in site energy production collectively, which is equivalent to about USD 300,000 in annual revenue losses. Among these, string faults were identified as the most significant contributor, accounting for more than 2,700 individual anomalies and resulting in a 3.48% decrease in total site production, equivalent to about USD 280,000 in annual revenue loss. These findings highlight the substantial financial and operational impact of undetected or misdiagnosed string-level faults in large-scale PV systems (Raptor Maps, 2025).

Furthermore, power losses from string faults highlight the ongoing challenge of keeping large PV systems running efficiently. Underperformance is a major cause due to these faults, which also presents clear opportunities to improve system output when identified and fixed. Recent data show that string-related faults increased by about 19% from 2023 to 2024, underlining how common this issue is and the need for better monitoring and fault detection in utility-scale PV systems (Raptor Maps, 2025).

3 Problem Definition

Solar energy plays a pivotal role in the global shift toward sustainable power generation. As large-scale PV systems increasingly contribute to electricity grids, it is essential in ensuring their consistent performance. However, many PV installations still rely on manual inspection methods to detect faults and inefficiencies. These manual processes are not only time-consuming, but they are also susceptible to human error (Abhishek Jha, Yogesh Rawat, Shruti Vyas, 2025). As PV systems grow larger and more complex, it becomes even more difficult to identify issues early, resulting in extended downtimes and unnecessary energy losses.

Undetected or late-detected faults can significantly affect both the output and the financial performance of solar plants. The lack of an automated, integrated solution means that operators often struggle to quickly localise and assess faults, leading to inefficient maintainance and increased downtime (Vinay Singh & Ruby Beniwal, 2025). Without smarter, data-driven systems, the true potential of solar energy cannot be fully realised.

In summary, there is an urgent need for advanced solutions that automate PV fault detection and diagnosis. Using recent advancements in Artificial Intelligence and integrating both electrical and image-based monitoring, operators can quickly identify, localise, and address faults, ultimately increasing system efficiency and supporting a more sustainable energy future.

4 Problem Statement

Inability of manual inspection to detect faults quickly and accurately in large solar PV systems leads to downtime and performance loss.

5 Research Motivation

Solar PV systems are very important as they help combat against climate change by producing clean, renewable electricity without greenhouse gas emissions. Yet, large-scale PV systems face downtime, energy losses and financial setbacks due to the electrical faults. By developing this AI-driven system, we hope to make solar PV operations more reliable and efficient.

6 Existing Work

Citation	Technology/Algorithm	Advantages	Limitations
(Li, B. et al., 2021)	Artificial Neural Network in combo with GADF	Robust to noise, captures full information	Dimensionality reduction had high computational cost and complexity.
(Sridharan et al., 2023)	DenseNet-201 in combo with J48 and WiSARD classifiers	Computationally efficient	Cannot detect electrical faults.
(Zhang et al., 2022)	NPCNN with WT, KNN and Isolation Forest	Improves accuracy. Handles outliers and nonlinearity.	Uses only historical data (no weather inputs)
(Mustafa et al., 2023)	Bi-LSTM	Great generalization with near perfect accuracy	Limited to three fault types.
(Dhingra, B., Tyagi, S. and Tomar, A., 2022)	Extra Tree regressor	Predicts PV degradation rate values closest to the actual values	Uses scientific/engineering degradation numeric data instead of electrical and meteorological data

7 Contribution to Body of Knowledge

7.1 Technological Contribution

Our application will be designed to automate and optimize photovoltaic (PV) system maintenance. Each four core components will leverage specific machine learning methodologies. For example, real-time fault detection will use supervised learning algorithms like Artificial Neural Networks and CNNs to identify anomalies (Li et al., 2021). Localizing these faults will be implemented using deep learning techniques, such as CNNs for image data and LSTMs for time-series electrical data (Mustafa et al., 2023)., to detect the exact location of the fault. The severity analysis will use regression models such as Gradient Boosting (Farooq et al., 2025), to identify the criticality and the potential impact of each fault. Finally, for fault rectification, a knowledge-based AI and predictive maintenance framework will recommend corrective actions for the faults that are identified.

7.2 Contribution to Domain

This research makes a pivotal contribution to the after-sales maintenance of solar farms by introducing an integrated intelligent diagnostic system that transforms maintenance from reactive to strategic, proactive, and predictive operations. Current practices lead to prolonged energy loss and unnecessary operational expenditures as critical faults go undetecteda challenge comprehensively documented in recent literature (Venkatakrishnan et al., 2023). Our system solves this by continuously identifying performance anomalies through automated fault detection. Once a fault is found, manually locating its source is slow and difficult (Ji et al., 2017); we address this with a precise fault location that determines the exact module or string. Even when located, operators often lack clarity on the fault's impact and how to prioritize it; we resolve this with a severity assessment that quantifies the energy loss and criticality. Finally, determining the best repair action causes delays; we close this gap with automated rectification, which recommends a solution to the faults, aiding in minimizing downtime and maximizing the energy yield.

8 Research Challenges

1. **Data Availability**
   Large, high-quality datasets containing both electrical and visual images are rare. This limits the ability to train and validate robust fault detection models. Therefore, we plan to combine public datasets and partner-collected data and simulate for better system predictions.

2. **Multi-modal Fusion**
   Synchronizing and combining heterogeneous data sources (electrical and visual) is non-trivial. To address this, we plan to design robust data fusion pipelines capable of aligning and integrating both modalities effectively.

3. **Environmental Variability**
   Variations in irradiance, temperature, and shading can alter signal patterns, leading to false positives. Developing adaptive models that can generalize under these dynamic environmental conditions is necessary.

4. **Designing a User-Friendly Interface**
   A simple and accessible interface is crucial for adoption by plant operators and maintenance teams. Therefore, dashboards presenting detection, localization, severity, and rectification guidance in an intuitive manner are essential. We will prototype web-based dashboards with clear visual overlays on site maps.

5. **Explainability, Operator Trust, and Safety**
   Before acting upon the suggestions, the field crew must be able to trust and understand them. Hence, Black-box outputs are likely to get rejected or unsafe actions. Therefore, the system must produce interpretable information to ensure operator confidence and safety.

9 Research Aim

The research aim is to design, develop, and evaluate an AI-driven system that enables solar PV operators to efficiently detect, locate, and assess faults, thereby providing actionable guidance to reduce operational downtime and energy losses.

10 Research Objectives

Objectives	Explanation	Learning Outcome
Problem Identification	PV system operators face major challenges in identifying and addressing faults early, causing performance loss and delayed maintenance.	LO1, LO4
Literature Review	**RO1:** Select and evaluate ML models for SCADA time-series data to detect anomalies and classify PV system faults. **RO2:** Identify computer vision models for drone thermal imagery to localize faults. **RO3:** Quantify fault severity and predict potential energy impact. **RO4:** Integrate detection, localization, and severity outputs to generate maintenance recommendations. **RO5:** Review existing diagnostic platforms and research to design a unified system for complete fault analysis and maintenance reports.	LO1, LO2, LO3, LO4
Data Gathering and Analysis	Our data collection uses industry-sourced SCADA telemetry, maintenance logs, and drone thermal images, with synthetic data added to address fault cases.	LO1, LO2, LO3, LO4
Research Design	This project will adopt a Design Science Research (DSR) paradigm. The core artifact will be built using a hybrid modeling approach, integrating algorithms such as Random Forest, Ensemble methods, and Neural Networks.	LO1, LO2, LO3
Implementation	• Implementation of Fault Detection by developing ML models to classify the operational state of the PV system and identify anomaly types. • Fault Localization using CNNs on thermal/EL imagery and LSTM for electrical signal analysis to pinpoint the exact module or string. • Prognostic severity model to quantify criticality by predicting power loss and potential degradation. • Recommendation engine translating outputs into actionable maintenance instructions.	LO2, LO3

Testing and Evaluation	Feedback from stakeholders, such as solar farm operators, maintenance managers, and field technicians, will be gathered through surveys and semi-structured interviews. This feedback will be used to evaluate the system's practical utility, the actionability of its diagnostics, and its effectiveness in streamlining maintenance workflows and reducing operational costs.	LO2, LO4

11 Project Scope

11.1 In-scope

No	Description
1	Building models to identify minor to moderate solar panel faults.
2	Evaluating system performance based on technical metrics (fault detection accuracy, localization, reduction in diagnostic time)
3	The focus is on proving the concept, not live deployment. Developing and validating the system using historical, batched datasets and synthetic data.
4	Generating fault reports with type of fault, its location, severity and recommended action

11.2 Out-scope

No	Description
1	Modification and physical repair to the solar PV system.
2	Does not extend to wind, hydro, or other renewable sources
3	Return on investments or financial cost modeling
4	Identifying catastrophic failures or micro-anomalies

12 Feature Prototype

Component 01 - Fault Detection

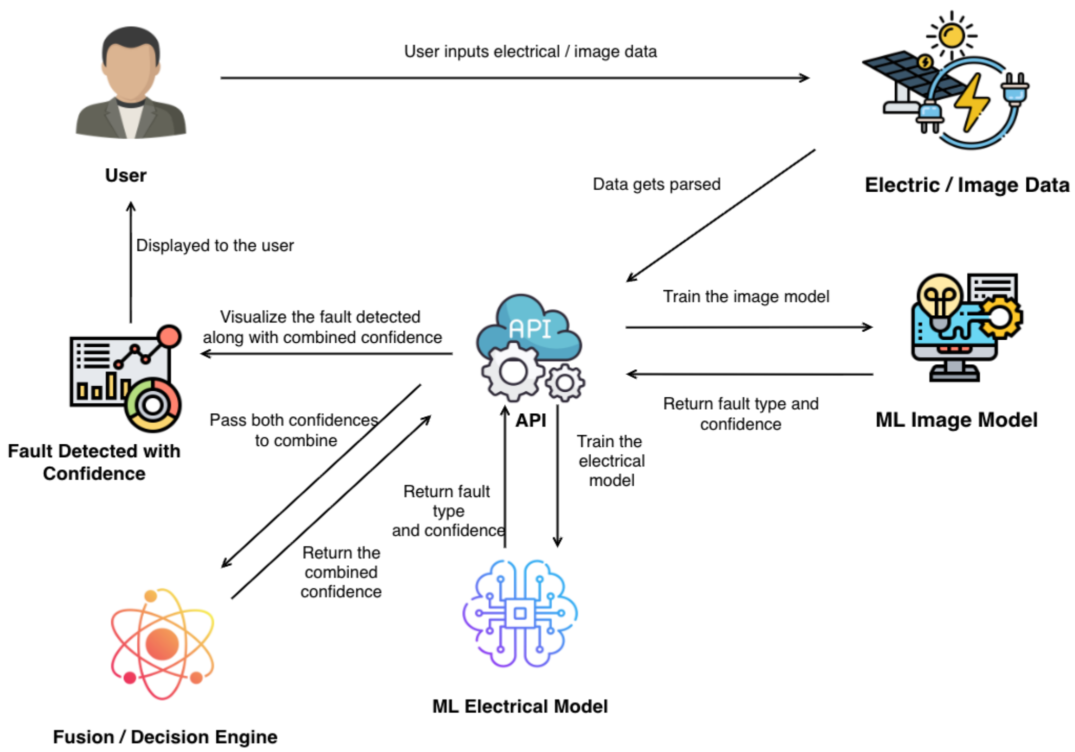
- Uploading Data - Electrical and image data in supported formats (CSV for electrical, JPG/PNG for images) containing I-V curves, string measurements, thermal images, and Electroluminescence imagery.

- Automated Data Processing - The system will process the data using signal transformation (GADF for I-V curves) and deep learning models such as ANN and DenseNet-201 to check for patterns and detect anomalies.

- Identify Fault Types - Faults are then classified into specific categories such as open-circuit, short-circuit, partial shading, and hotspots based on electrical and visual evidence.

- Display Detection Results - Fault detection results are displayed to the user showing the fault type and the detection confidence score for each identified issue.

- Plan initial response - Based on the faults detected and the confidence, the user can decide the required actions needed to handle these faults.



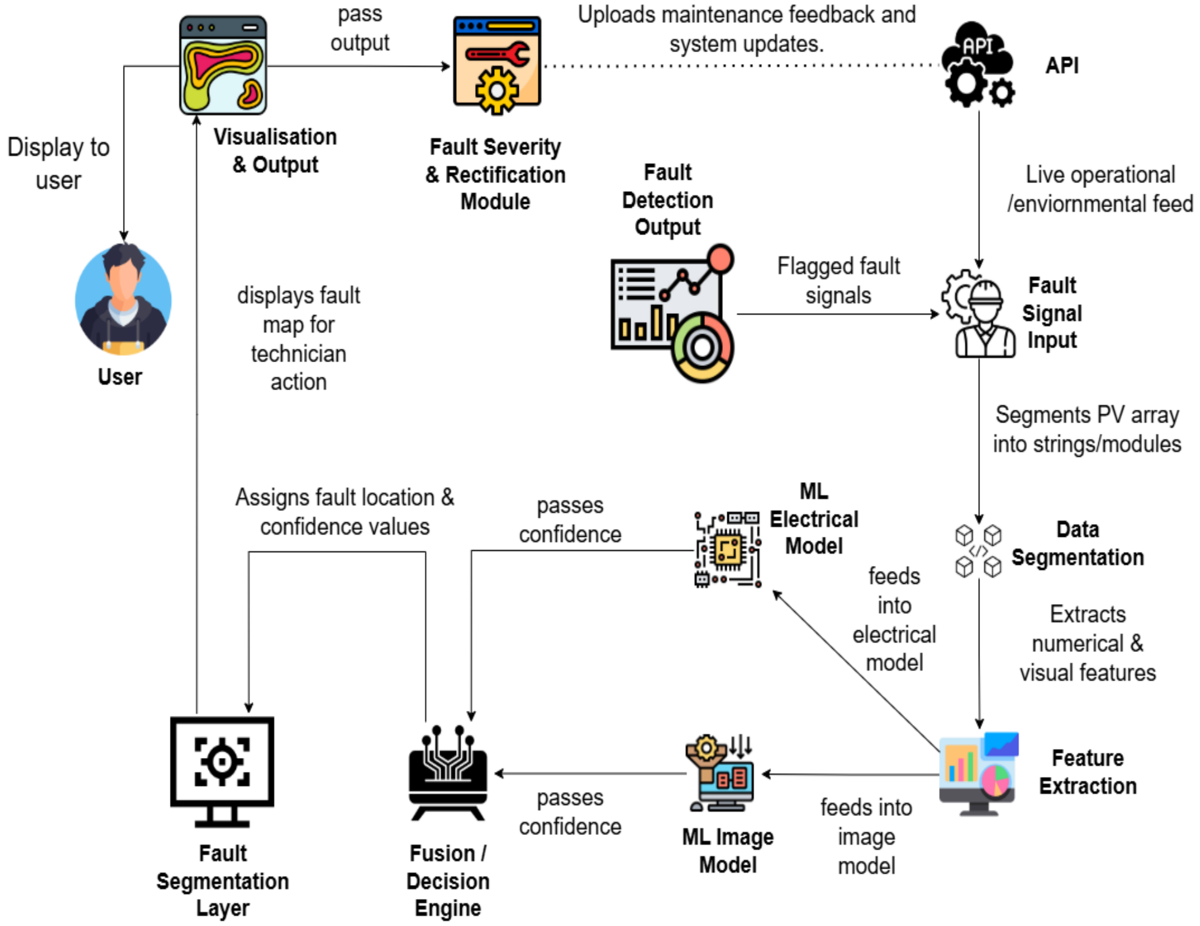
Component 02 - Fault Localisation

- Upload fault Data and images - User uploads the isolated data(current, voltage) from the detected fault, along with available thermal or electroluminescence imagery of the affected PV array sections.

- Segment Array and Data - The PV array is logically divided into segments (strings and modules). The Data from sensors and images will be processed and grouped according to these segments for individual analysis.

- Extract Electrical and Visual Features - The system will automatically extract numerical features from the time-series data and visual features from the images that are relevant to the fault and identify any anomalies.

- Run Localization Models - The Extracted features are processed by an electrical ML model that will analyse patterns in the sensor data and an image ML model that will analyse the imagery.
- Fuse Results and Generate Fault Map - The fusion engine will combine the outputs from the two machine learning models (electrical and image models) to confirm the fault's precise location. The system then generates a spatial fault map.
- View localized Fault - An interactive visualization of the fault and its location will be displayed to the user. Highlighting the faulty strings or modules, for quick identification and action.

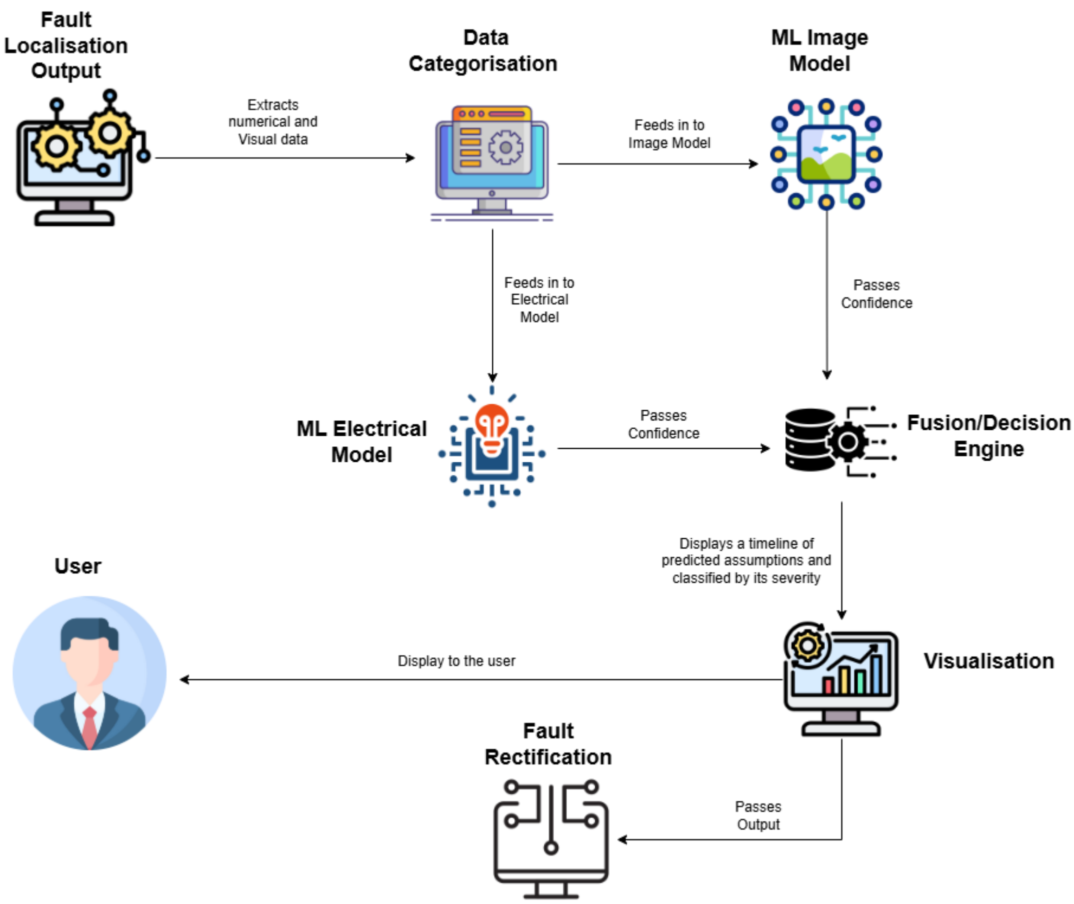


12.1 Component 03 - Fault Severity

- Input Data - The system receives a CSV file containing both electrical and image data from the Fault Localization component. This dataset includes key operational parameters and corresponding visual data for each affected module or string.
- Fault Categorization – The uploaded dataset is automatically organized based on data type and fault category before being processed with the appropriate analytical model.
- Severity Prediction – The electrical data is analyzed using regression-based machine learning models such as the Decision Tree Regressor to predict the level of degradation, power loss, and estimated repair costs. Simultaneously, the image data is processed using

a deep learning model designed to analyze and classify the severity of physical damage observed in the components.

- Classification and Visualization of Results – The final results are displayed through interactive charts and tables that visualize degradation patterns, fault severity levels, and their associated economic impact.



12.2 Component 04 - Fault Rectification

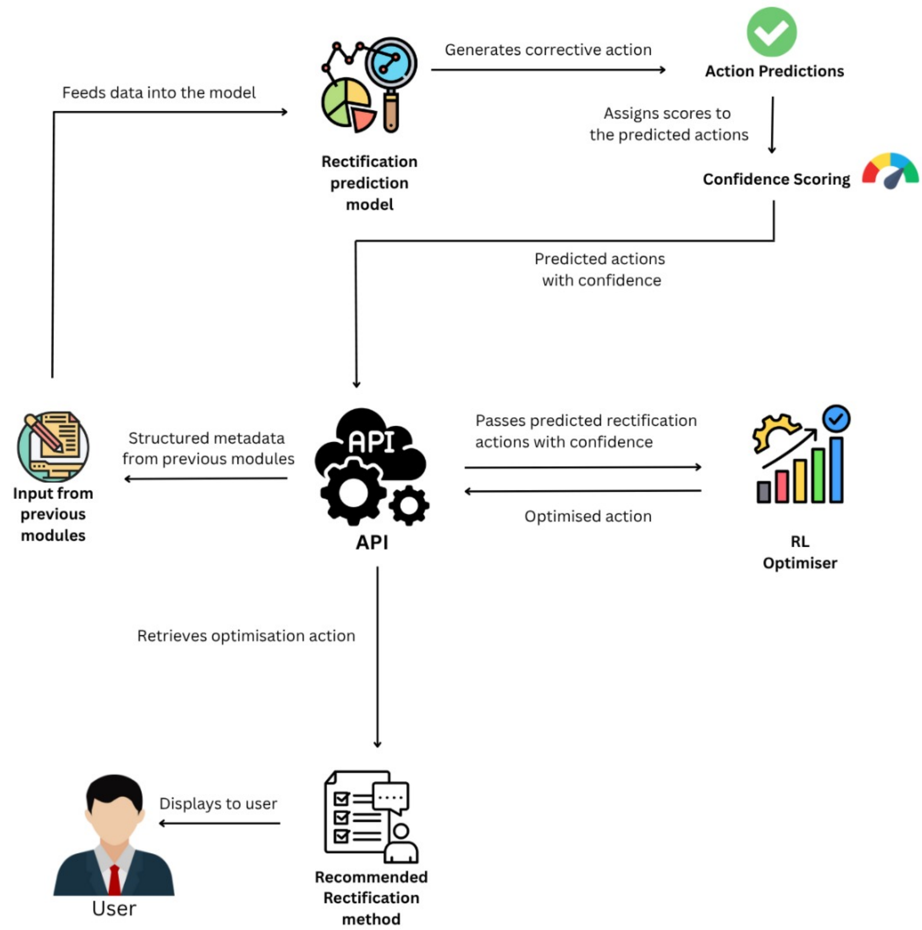
- Input Data – The system receives detailed fault data including type, location, severity environmental factors and maintenance history from the upper modules.

- Rectification Prediction Model – Using the confidence scores and fault data received from upstream by using a machine learning model like Random Forest or ANN the model predicts the corrective actions along with the confidence scores and estimated outcome.

- RL Optimiser – The Deep Q-Network RL agent receives the predicted rectification actions along with the confidence scores and optimises the final recommendation strategy based on cost, downtime, restoration speed and reliability.

- Recommended Rectification Method – The optimised action from the RL optimizer is structured into a detailed recommendation containing corrective action, implementation steps, timeframe, required resources expected performance gains, cost and confidence score.

- User Display Interface – The final recommendation is displayed to the user via an UI interface.



13 Methodology

13.1 Research Methodology

Research Philosophy	We will adopt a pragmatist Philosophy, as it prioritizes the practical problem solving nature of our research. This approach legitimizes the synthesis of hard, quantitative data and soft qualitative knowledge.
Research Approach	A deductive approach will be utilized. This research uses this approach to test specific machine learning models for PV fault tasks.
Research Strategy	The research employs a multi-strategy design using Design Science Research as the primary methodology, supplemented by case studies for system validation and simulations for robustness testing.
Research Choice	A mixed-methods approach combines quantitative performance metrics (accuracy, precision, MAE) with qualitative feedback from maintenance engineers to evaluate both technical efficacy and practical utility.
Time Zone	The project uses a cross-sectional approach for core model validation on historical data, while incorporating longitudinal elements for prognostic severity prediction and continuous learning.

13.2 Development Methodology

13.2.1 Lifecycle Model

We plan to use the Agile life cycle model due to its continuous delivery and iterative development through sprints. Each sprint helps us develop, test, and improve the components of our AI-driven solar PV fault management system, while having the ability to respond quickly to new changes and feedback to ensure a reliable and high-quality solution.

13.2.2 Design Methodology

This project will follow a hybrid approach combining both SSADM and OOAD. SSADM provides the ability to map the data and process flows (SCADA, images, preprocessing), ensuring clear requirements and data quality. OOAD will convert these flows into modular software components. Hence, this makes development, testing, and scaling easier and reduces integration risk.

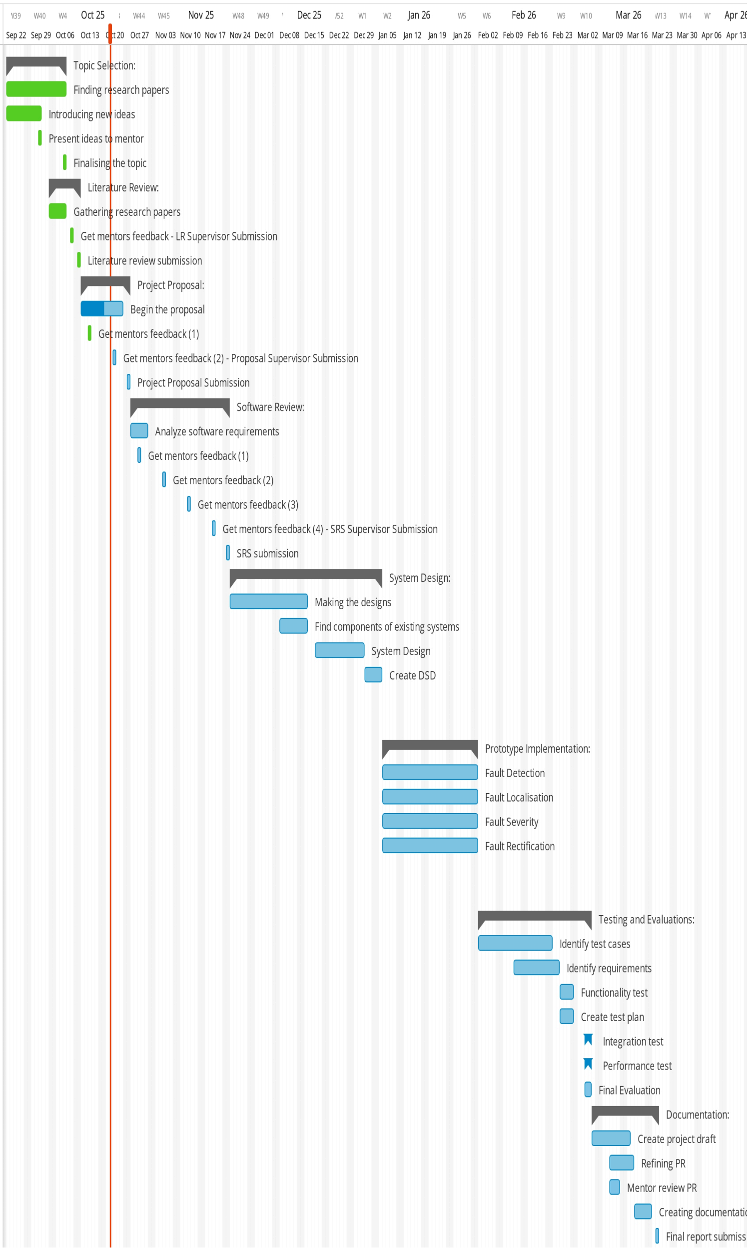
13.2.3 Evaluation Methodology

The system will be evaluated using standard metrics (accuracy, precision, recall, F1-score) for classification tasks. The Mean Absolute Error (MAE) and Root Mean Square Error (RMSE) will be used to access the performance of the severity analysis model. The performance will be benchmarked against industry models

13.3 Project Management Methodology

This project makes use of the Scrum framework in the Agile methodology. The development process is divided into short sprints, where each sprint focuses on specific deliverables such as data collection, model development, and interface design. The team will hold regular meetings to review progress, identify possible challenges, and plan the next sprint. This iterative and collaborative approach ensures the flexibility in adapting to changes, promotes clear communication between members, and efficient achievement of project milestones.

13.3.1 Gantt Chart



Solar PV Fault Localisation and Rectification

Read-only view, generated on 21 Oct 2025

Instagantt

	ACTIVITIES	ASSIGNEE	EH	START	DUE	STATUS	%
	Topic Selection:		-	22/Sep	08/Oct		100%
1	Finding research papers	Unassigned	-	22/Sep	08/Oct	Finished	100%
2	Introducing new ideas	Unassigned	-	22/Sep	01/Oct	Finished	100%
3	Present ideas to mentor	Unassigned	-	01/Oct	01/Oct	Finished	100%
4	Finalising the topic	Unassigned	-	08/Oct	08/Oct	Approved	100%
	Literature Review:		-	04/Oct	12/Oct		100%
6	Gathering research papers	Unassigned	-	04/Oct	08/Oct	Finished	100%
7	Get mentors feedback - LR ...	Unassigned	-	10/Oct	10/Oct	Approved	100%
8	Literature review submission	Unassigned	-	12/Oct	12/Oct	Finished	100%
	Project Proposal:		-	13/Oct	26/Oct		50%
10	Begin the proposal	Unassigned	-	13/Oct	24/Oct	Started	55%
11	Get mentors feedback (1)	Unassigned	-	15/Oct	15/Oct	Finished	100%
12	Get mentors feedback (2) - ...	Unassigned	-	22/Oct	22/Oct	Approval reques	0%
13	Project Proposal Submission	Unassigned	-	26/Oct	26/Oct	Started	0%
	Software Review:		-	27/Oct	23/Nov		0%
15	Analyze software requirem...	Unassigned	-	27/Oct	31/Oct	Not started	0%
16	Get mentors feedback (1)	Unassigned	-	29/Oct	29/Oct	Not started	0%
17	Get mentors feedback (2)	Unassigned	-	05/Nov	05/Nov	Not started	0%
18	Get mentors feedback (3)	Unassigned	-	12/Nov	12/Nov	Not started	0%
19	Get mentors feedback (4) - ...	Unassigned	-	19/Nov	19/Nov	Not started	0%
20	SRS submission	Unassigned	-	23/Nov	23/Nov	Not started	0%
	System Design:		-	24/Nov	05/Jan		0%
22	Making the designs	Unassigned	-	24/Nov	15/Dec	Not started	0%
23	Find components of existing...	Unassigned	-	08/Dec	15/Dec	Not started	0%
24	System Design	Unassigned	-	18/Dec	31/Dec	Not started	0%
25	Create DSD	Unassigned	-	01/Jan	05/Jan	Not started	0%
26	Mentor review DSD	Unassigned	-			Not started	0%
27	Self evaluate DSD	Unassigned	-			Not started	0%
	Prototype Implementation:		-	06/Jan	01/Feb		0%
29	Fault Detection	Unassigned	-	06/Jan	01/Feb	Not started	0%
30	Fault Localisation	Unassigned	-	06/Jan	01/Feb	Not started	0%
31	Fault Severity	Unassigned	-	06/Jan	01/Feb	Not started	0%
32	Fault Rectification	Unassigned	-	06/Jan	01/Feb	Not started	0%
33	Demonstrate prototype	Unassigned	-			Not started	0%
34	Refine prototype	Unassigned	-			Not started	0%
	Testing and Evaluations:		-	02/Feb	05/Mar		0%
36	Identify test cases	Unassigned	-	02/Feb	22/Feb	Not started	0%
37	Identify requirements	Unassigned	-	12/Feb	24/Feb	Not started	0%
38	Functionality test	Unassigned	-	25/Feb	28/Feb	Not started	0%
39	Create test plan	Unassigned	-	25/Feb	28/Feb	Not started	0%
	Integration test	Unassigned	-	04/Mar	05/Mar	Not started	0%
	Performance test	Unassigned	-	04/Mar	05/Mar	Not started	0%
42	Final Evaluation	Unassigned	-	04/Mar	05/Mar	Not started	0%
	Documentation:		-	06/Mar	24/Mar		0%
44	Create project draft	Unassigned	-	06/Mar	16/Mar	Not started	0%
45	Refining PR	Unassigned	-	11/Mar	17/Mar	Not started	0%
46	Mentor review PR	Unassigned	-	11/Mar	13/Mar	Not started	0%
47	Creating documentation	Unassigned	-	18/Mar	22/Mar	Not started	0%
48	Final report submission	Unassigned	-	24/Mar	24/Mar	Not started	0%

14 Deliverables

Deliverable	Date / Week
Semester 01	
Submission of Literature Review	Week 03
Submission of Project Proposal to the supervisor	Week 04
Final Project Proposal submission	Week 05
SRS submission to the supervisor	Week 08
Final SRS submission	Week 09
Semester 02	
Prototype Implementation	Week 14
Testing and Evaluation	Week 19
Documentation and final report submission	Week 23

15 Resource Requirements

15.1 Hardware Requirements

• **CPU with good performance** (Intel Core i7 9th gen/Ryzen 7 processor or better)- Capable of handling resource-intensive tasks and demanding computer-based operations.
• **16GB RAM or higher** - Ensures efficient processing of large datasets, running algorithms, and training machine learning models.
• **Atleast 250GB SSD or 1TB HDD**: Provides sufficient storage for datasets, application code, and testing files.

15.2 Software Requirements

Python: Used for AI/ML development, data analysis, and integration.

Scikit-learn: Provides machine learning algorithms, evaluation, and pre-processing tools.

TensorFlow, PyTorch: For ML models using image processing

Google Colab: Cloud-based platform for running Python notebooks and collaborative coding.

PyCharm: To test and debug code

LaTeX / Microsoft Word/ Notion: Used for writing reports, proposal, and documentation.

Windows OS / Mac OS: Operating systems used for high-performance tasks.

Streamlit: For the final output dashboard displayed

FastAPI: To act as the bridge for the frontend and backend

15.3 Data Requirements

Our model development will primarily focus on building a highly accurate core model by using directly targeted proprietary datasets from partner solar farms. This approach ensures that we train our models to learn from real-world operational complexities. We will utilize techniques such as transfer learning and synthetic data generation to adapt to a wider scale of solutions for the range of PV system configurations and geographical settings.

15.4 Skill Requirements

- Data Analysis and Visualization
- Time Management and Planning
- Critical Thinking
- Data Collection and Preprocessing
- Technical Writing
- Programming and Implementation
- UI/UX Design
- Communication and team Collaboration
- Research Skills

16 Risk Management

Risk Item	Severity	Frequency	Mitigation Plan
Data unavailability	5	5	Public datasets along with partner-collected data are combined, along with synthetic data.
Model underperformance	5	4	Multiple ML algorithms are tested, hyperparameter tuning is performed, and validation is done using cross-validation.
Hardware or software failure	5	4	Backups must be maintained, cloud platforms like Google Colab, and Github are used to ensure regular system updates.
User Misinterpretation of Results	5	4	Clear visualizations and explanations should be provided.

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